CycleGAN

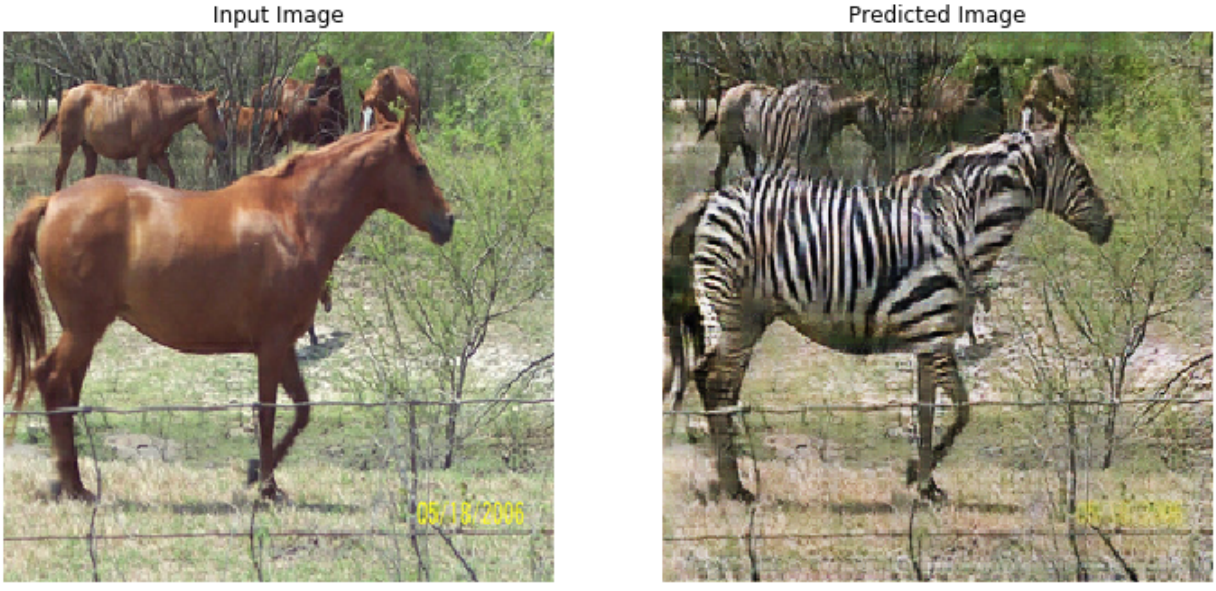
Run in Google Colab (https://colab.research.google.com/github/tensorflow/docs/blob/master/site/en/tutorials

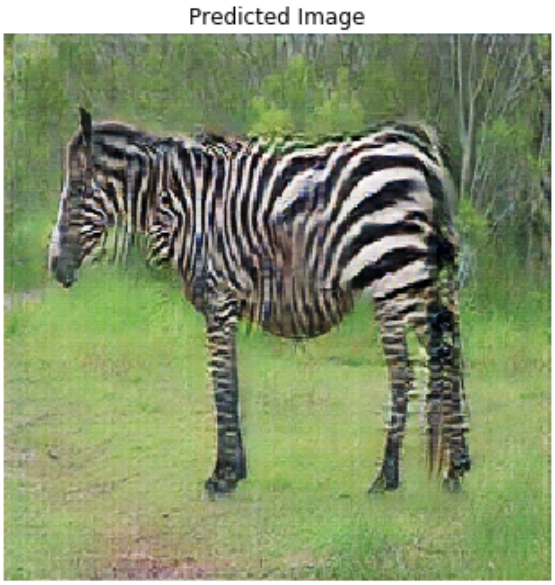
This notebook demonstrates unpaired image to image translation using conditional GAN's, as described in Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks (https://arxiv.org/abs/1703.10593), also known as CycleGAN. The paper proposes a method that can capture the characteristics of one image domain and figure out how these characteristics could be translated into another image domain, all in the absence of any paired training examples.

This notebook assumes you are familiar with Pix2Pix, which you can learn about in the Pix2Pix tutorial (https://www.tensorflow.org/tutorials/generative/pix2pix). The code for CycleGAN is similar, the main difference is an additional loss function, and the use of unpaired training data.

CycleGAN uses a cycle consistency loss to enable training without the need for paired data. In other words, it can translate from one domain to another without a one-to-one mapping between the source and target domain.

This opens up the possibility to do a lot of interesting tasks like photo-enhancement, image colorization, style transfer, etc. All you need is the source and the target dataset (which is simply a directory of images).





Set up the input pipeline

Install the tensorflow_examples (https://github.com/tensorflow/examples) package that enables importing of the generator and the discriminator.

```
$ pip install git+https://github.com/tensorflow/examples.git
```

```
import tensorflow as tf
```

```
2022-12-14 05:17:07.970694: W tensorflow/compiler/xla/stream_executor/platfor
2022-12-14 05:17:07.970801: W tensorflow/compiler/xla/stream_executor/platfor
2022-12-14 05:17:07.970810: W tensorflow/compiler/tf2tensorrt/utils/py_utils.
```

```
import tensorflow_datasets as tfds
from tensorflow_examples.models.pix2pix import pix2pix

import os
import time
import matplotlib.pyplot as plt
from IPython.display import clear_output
```

```
AUTOTUNE = tf.data.AUTOTUNE
```

Input Pipeline

This tutorial trains a model to translate from images of horses, to images of zebras. You can find this dataset and similar ones here (https://www.tensorflow.org/datasets/catalog/cycle_gan).

As mentioned in the paper (https://arxiv.org/abs/1703.10593), apply random jittering and mirroring to the training dataset. These are some of the image augmentation techniques that avoids overfitting.

This is similar to what was done in pix2pix (https://www.tensorflow.org/tutorials/generative/pix2pix#load_the_dataset)

- In random jittering, the image is resized to `286 x 286` and then randomly cropped to `256 x 256`.
- In random mirroring, the image is randomly flipped horizontally i.e. left to right.

```
dataset, metadata = tfds.load('cycle_gan/horse2zebra',
                              with_info=True, as_supervised=True)

train_horses, train_zebras = dataset['trainA'], dataset['trainB']
test_horses, test_zebras = dataset['testA'], dataset['testB']
```

```
BUFFER_SIZE = 1000
BATCH_SIZE = 1
IMG_WIDTH = 256
IMG_HEIGHT = 256
```

```
def random_crop(image):
  cropped_image = tf.image.random_crop(
      image, size=[IMG_HEIGHT, IMG_WIDTH, 3])

  return cropped_image
```

```
# normalizing the images to [-1, 1]
def normalize(image):
  image = tf.cast(image, tf.float32)
  image = (image / 127.5) - 1
  return image



def random_jitter(image):
  # resizing to 286 x 286 x 3
  image = tf.image.resize(image, [286, 286],
                          method=tf.image.ResizeMethod.NEAREST_NEIGHBOR)

  # randomly cropping to 256 x 256 x 3
  image = random_crop(image)

  # random mirroring
  image = tf.image.random_flip_left_right(image)

  return image



def preprocess_image_train(image, label):
  image = random_jitter(image)
  image = normalize(image)
  return image



def preprocess_image_test(image, label):
  image = normalize(image)
  return image



train_horses = train_horses.cache().map(
    preprocess_image_train, num_parallel_calls=AUTOTUNE).shuffle(
    BUFFER_SIZE).batch(BATCH_SIZE)

train_zebras = train_zebras.cache().map(
    preprocess_image_train, num_parallel_calls=AUTOTUNE).shuffle(
    BUFFER_SIZE).batch(BATCH_SIZE)

test_horses = test_horses.map(
    preprocess_image_test, num_parallel_calls=AUTOTUNE).cache().shuffle(
```

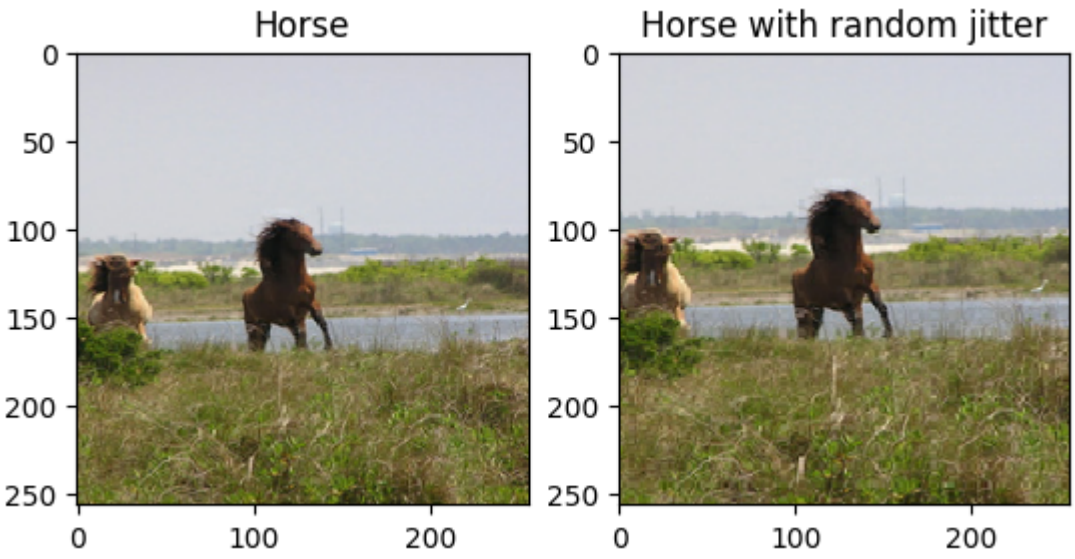
```
    BUFFER_SIZE).batch(BATCH_SIZE)

test_zebras = test_zebras.map(
    preprocess_image_test, num_parallel_calls=AUTOTUNE).cache().shuffle(
    BUFFER_SIZE).batch(BATCH_SIZE)
```

```
sample_horse = next(iter(train_horses))
sample_zebra = next(iter(train_zebras))
```

```
2022-12-14 05:17:15.849959: W tensorflow/core/kernels/data/cache_dataset_ops.
2022-12-14 05:17:16.911316: W tensorflow/core/kernels/data/cache_dataset_ops.
```

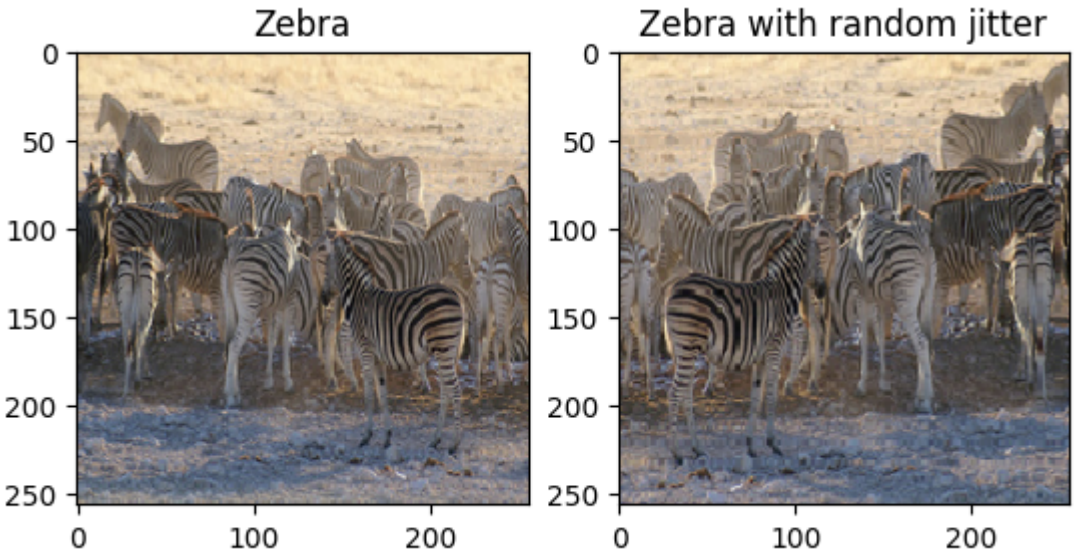
```
plt.subplot(121)
plt.title('Horse')
plt.imshow(sample_horse[0] * 0.5 + 0.5)

plt.subplot(122)
plt.title('Horse with random jitter')
plt.imshow(random_jitter(sample_horse[0]) * 0.5 + 0.5)
```

```
<matplotlib.image.AxesImage at 0x7f48443caf40>
```



```
plt.subplot(121)
plt.title('Zebra')
plt.imshow(sample_zebra[0] * 0.5 + 0.5)

plt.subplot(122)
plt.title('Zebra with random jitter')
plt.imshow(random_jitter(sample_zebra[0]) * 0.5 + 0.5)
```

```
<matplotlib.image.AxesImage at 0x7f469874e550>
```



Import and reuse the Pix2Pix models

Import the generator and the discriminator used in Pix2Pix (https://github.com/tensorflow/examples/blob/master/tensorflow_examples/models/pix2pix/pix2pix.py)
via the installed tensorflow_examples (https://github.com/tensorflow/examples) package.

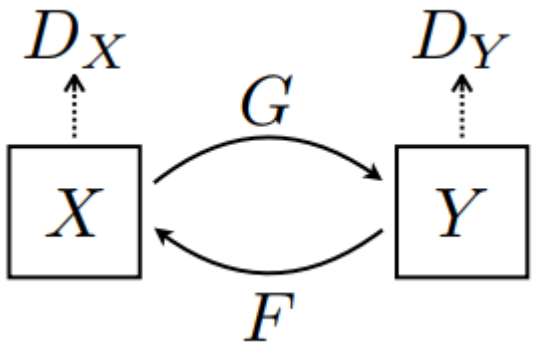
The model architecture used in this tutorial is very similar to what was used in pix2pix (https://github.com/tensorflow/examples/blob/master/tensorflow_examples/models/pix2pix/pix2pix.py)
. Some of the differences are:

- Cyclegan uses instance normalization (https://arxiv.org/abs/1607.08022) instead of batch normalization (https://arxiv.org/abs/1502.03167).

- The CycleGAN paper (https://arxiv.org/abs/1703.10593) uses a modified `resnet` based generator. This tutorial is using a modified `unet` generator for simplicity.

There are 2 generators (G and F) and 2 discriminators (X and Y) being trained here.

- Generator `G` learns to transform image `X` to image `Y`. $(G : X -> Y)$
- Generator `F` learns to transform image `Y` to image `X`. $(F : Y -> X)$
- Discriminator `D_X` learns to differentiate between image `X` and generated image `X` (`F(Y)`).
- Discriminator `D_Y` learns to differentiate between image `Y` and generated image `Y` (`G(X)`).

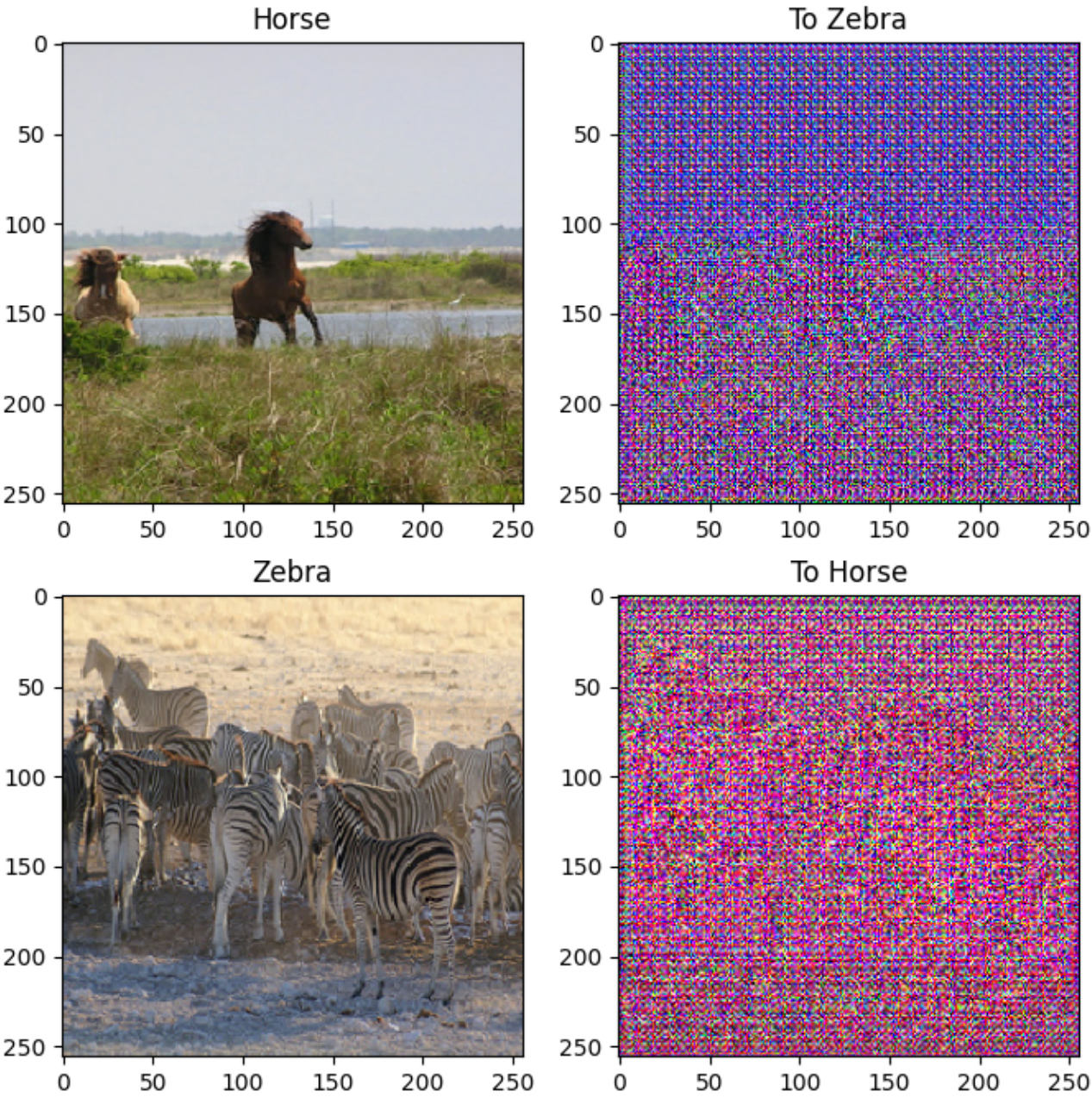



```
OUTPUT_CHANNELS = 3

generator_g = pix2pix.unet_generator(OUTPUT_CHANNELS, norm_type='instancenorm
generator_f = pix2pix.unet_generator(OUTPUT_CHANNELS, norm_type='instancenorm

discriminator_x = pix2pix.discriminator(norm_type='instancenorm', target=Fals
discriminator_y = pix2pix.discriminator(norm_type='instancenorm', target=Fals
```

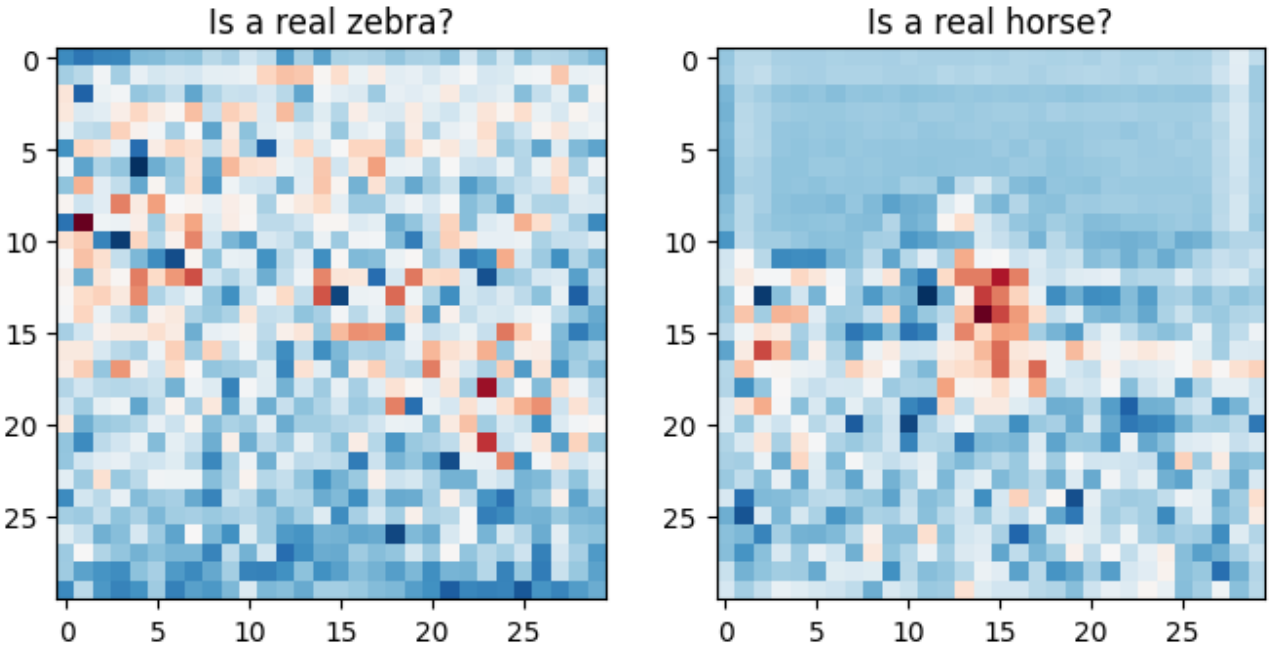
```
to_zebra = generator_g(sample_horse)
to_horse = generator_f(sample_zebra)
plt.figure(figsize=(8, 8))
contrast = 8

imgs = [sample_horse, to_zebra, sample_zebra, to_horse]
title = ['Horse', 'To Zebra', 'Zebra', 'To Horse']

for i in range(len(imgs)):
  plt.subplot(2, 2, i+1)
```

```
  plt.title(title[i])
  if i % 2 == 0:
    plt.imshow(imgs[i][0] * 0.5 + 0.5)
  else:
    plt.imshow(imgs[i][0] * 0.5 * contrast + 0.5)
plt.show()
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow wi
WARNING:matplotlib.image:Clipping input data to the valid range for imshow wi
```



```
plt.figure(figsize=(8, 8))

plt.subplot(121)
plt.title('Is a real zebra?')
plt.imshow(discriminator_y(sample_zebra)[0, ..., -1], cmap='RdBu_r')

plt.subplot(122)
plt.title('Is a real horse?')
plt.imshow(discriminator_x(sample_horse)[0, ..., -1], cmap='RdBu_r')

plt.show()
```



Loss functions

In CycleGAN, there is no paired data to train on, hence there is no guarantee that the input `x` and the target `y` pair are meaningful during training. Thus in order to enforce that the network learns the correct mapping, the authors propose the cycle consistency loss.

The discriminator loss and the generator loss are similar to the ones used in pix2pix (https://www.tensorflow.org/tutorials/generative/pix2pix#build_the_generator).

```
LAMBDA = 10
```

```
loss_obj = tf.keras.losses.BinaryCrossentropy(from_logits=True)
```

```
def discriminator_loss(real, generated):
  real_loss = loss_obj(tf.ones_like(real), real)

  generated_loss = loss_obj(tf.zeros_like(generated), generated)

  total_disc_loss = real_loss + generated_loss

  return total_disc_loss * 0.5
```

```
def generator_loss(generated):
  return loss_obj(tf.ones_like(generated), generated)
```

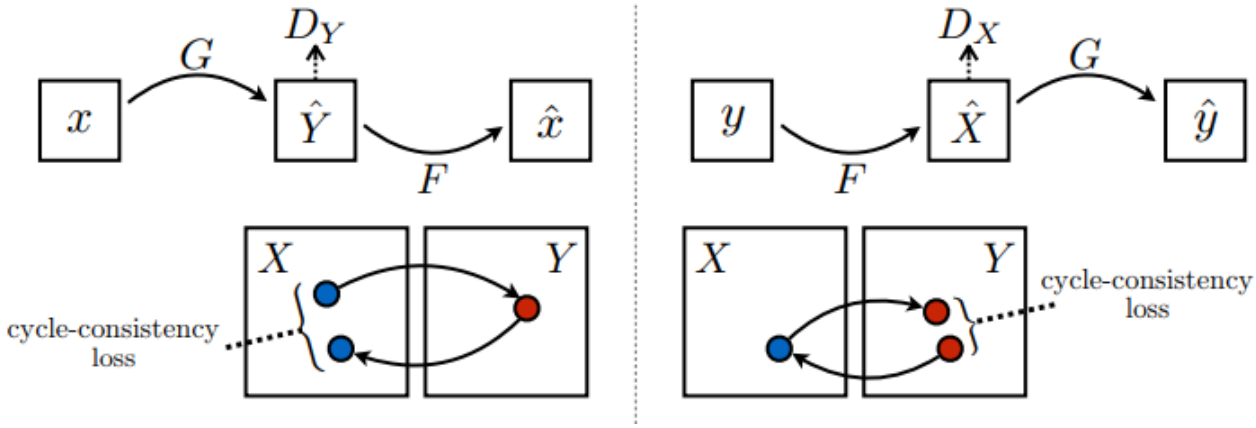
Cycle consistency means the result should be close to the original input. For example, if one translates a sentence from English to French, and then translates it back from French to English, then the resulting sentence should be the same as the original sentence.

In cycle consistency loss,

- Image X is passed via generator G that yields generated image $\hat{Y}$.
- Generated image $\hat{Y}$ is passed via generator F that yields cycled image $\hat{X}$.
- Mean absolute error is calculated between X and $\hat{X}$.

$$forward\ cycle\ consistency\ loss: X -> G(X) -> F(G(X)) \sim \hat{X}$$

$$backward\ cycle\ consistency\ loss: Y -> F(Y) -> G(F(Y)) \sim \hat{Y}$$



```
def calc_cycle_loss(real_image, cycled_image):
  loss1 = tf.reduce_mean(tf.abs(real_image - cycled_image))

  return LAMBDA * loss1
```

As shown above, generator G is responsible for translating image X to image Y. Identity loss says that, if you fed image Y to generator G, it should yield the real image Y or something close to image Y.

If you run the zebra-to-horse model on a horse or the horse-to-zebra model on a zebra, it should not modify the image much since the image already contains the target class.

$$Identity\ loss = |G(Y) - Y| + |F(X) - X|$$

```
def identity_loss(real_image, same_image):
  loss = tf.reduce_mean(tf.abs(real_image - same_image))
  return LAMBDA * 0.5 * loss
```

Initialize the optimizers for all the generators and the discriminators.

```
generator_g_optimizer = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)
generator_f_optimizer = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)

discriminator_x_optimizer = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)
discriminator_y_optimizer = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)
```

Checkpoints

```
checkpoint_path = "./checkpoints/train"

ckpt = tf.train.Checkpoint(generator_g=generator_g,
                           generator_f=generator_f,
                           discriminator_x=discriminator_x,
                           discriminator_y=discriminator_y,
                           generator_g_optimizer=generator_g_optimizer,
                           generator_f_optimizer=generator_f_optimizer,
                           discriminator_x_optimizer=discriminator_x_optimize
                           discriminator_y_optimizer=discriminator_y_optimize
```

```
ckpt_manager = tf.train.CheckpointManager(ckpt, checkpoint_path, max_to_keep=5

# if a checkpoint exists, restore the latest checkpoint.
if ckpt_manager.latest_checkpoint:
  ckpt.restore(ckpt_manager.latest_checkpoint)
  print ('Latest checkpoint restored!!')
```

Training

Note: This example model is trained for fewer epochs (10) than the paper (200) to keep training time reasonable for this tutorial. The generated images will have much lower quality.

```
EPOCHS = 10
```

```
def generate_images(model, test_input):
  prediction = model(test_input)

  plt.figure(figsize=(12, 12))

  display_list = [test_input[0], prediction[0]]
  title = ['Input Image', 'Predicted Image']

  for i in range(2):
    plt.subplot(1, 2, i+1)
    plt.title(title[i])
    # getting the pixel values between [0, 1] to plot it.
    plt.imshow(display_list[i] * 0.5 + 0.5)
    plt.axis('off')
  plt.show()
```

Even though the training loop looks complicated, it consists of four basic steps:

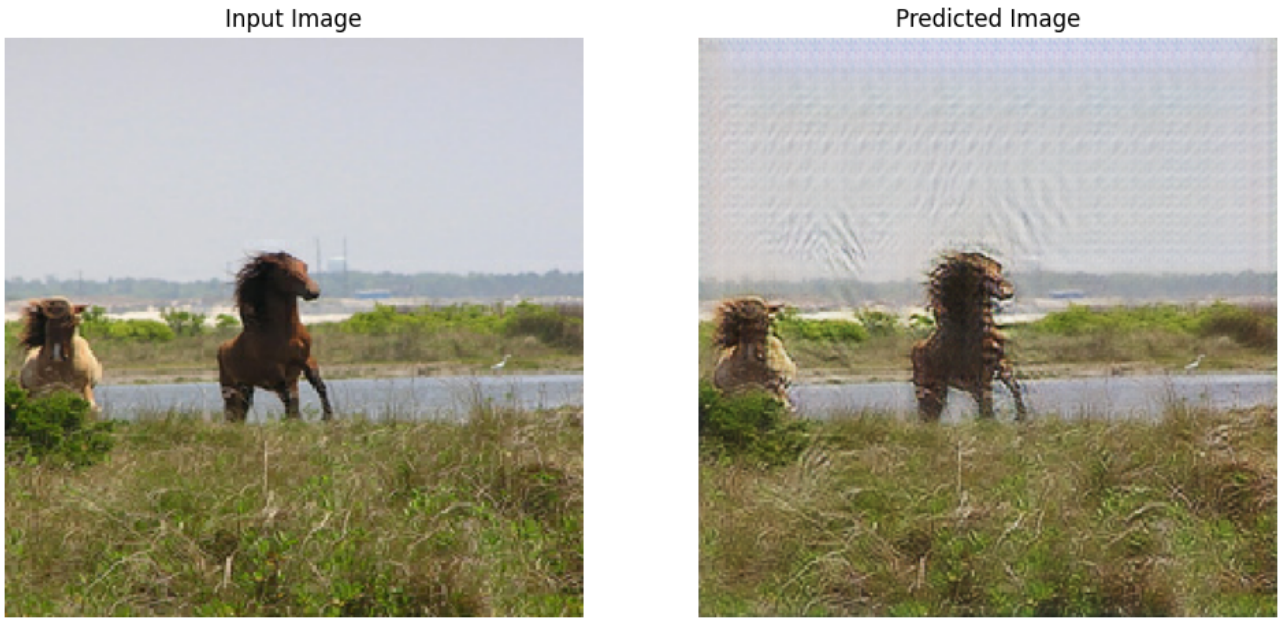
- Get the predictions.
- Calculate the loss.
- Calculate the gradients using backpropagation.

- Apply the gradients to the optimizer.

```
@tf.function
def train_step(real_x, real_y):
  # persistent is set to True because the tape is used more than
  # once to calculate the gradients.
  with tf.GradientTape(persistent=True) as tape:
    # Generator G translates X -> Y
    # Generator F translates Y -> X.

    fake_y = generator_g(real_x, training=True)
    cycled_x = generator_f(fake_y, training=True)

    fake_x = generator_f(real_y, training=True)
    cycled_y = generator_g(fake_x, training=True)

    # same_x and same_y are used for identity loss.
    same_x = generator_f(real_x, training=True)
    same_y = generator_g(real_y, training=True)

    disc_real_x = discriminator_x(real_x, training=True)
    disc_real_y = discriminator_y(real_y, training=True)

    disc_fake_x = discriminator_x(fake_x, training=True)
    disc_fake_y = discriminator_y(fake_y, training=True)

    # calculate the loss
    gen_g_loss = generator_loss(disc_fake_y)
    gen_f_loss = generator_loss(disc_fake_x)

    total_cycle_loss = calc_cycle_loss(real_x, cycled_x) + calc_cycle_loss(rea

    # Total generator loss = adversarial loss + cycle loss
    total_gen_g_loss = gen_g_loss + total_cycle_loss + identity_loss(real_y, s
    total_gen_f_loss = gen_f_loss + total_cycle_loss + identity_loss(real_x, s

    disc_x_loss = discriminator_loss(disc_real_x, disc_fake_x)
    disc_y_loss = discriminator_loss(disc_real_y, disc_fake_y)

  # Calculate the gradients for generator and discriminator
  generator_g_gradients = tape.gradient(total_gen_g_loss,
                                        generator_g.trainable_variables)
  generator_f_gradients = tape.gradient(total_gen_f_loss,
                                        generator_f.trainable_variables)

  discriminator_x_gradients = tape.gradient(disc_x_loss,
                                            discriminator_x.trainable_variable
```

```
    discriminator_y_gradients = tape.gradient(disc_y_loss,
                                              discriminator_y.trainable_variable

    # Apply the gradients to the optimizer
    generator_g_optimizer.apply_gradients(zip(generator_g_gradients,
                                              generator_g.trainable_variables))

    generator_f_optimizer.apply_gradients(zip(generator_f_gradients,
                                              generator_f.trainable_variables))

    discriminator_x_optimizer.apply_gradients(zip(discriminator_x_gradients,
                                                  discriminator_x.trainable_var

    discriminator_y_optimizer.apply_gradients(zip(discriminator_y_gradients,
                                                  discriminator_y.trainable_var
```

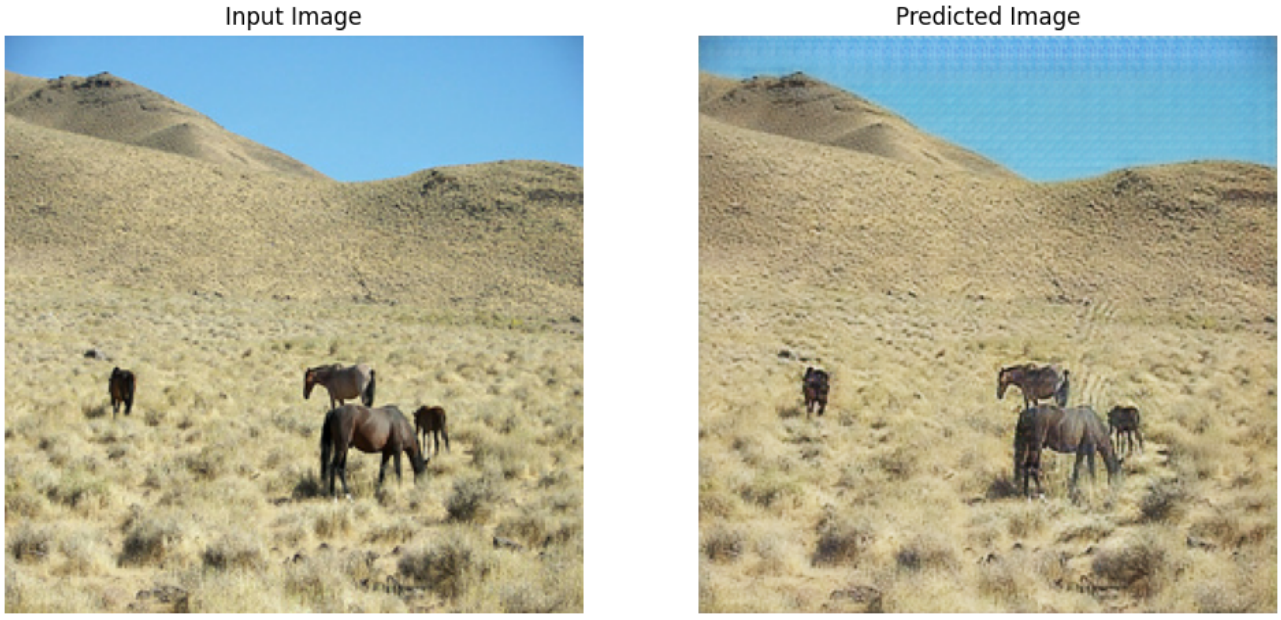
```
for epoch in range(EPOCHS):
  start = time.time()

  n = 0
  for image_x, image_y in tf.data.Dataset.zip((train_horses, train_zebras)):
    train_step(image_x, image_y)
    if n % 10 == 0:
      print ('.', end='')
    n += 1

  clear_output(wait=True)
  # Using a consistent image (sample_horse) so that the progress of the model
  # is clearly visible.
  generate_images(generator_g, sample_horse)

  if (epoch + 1) % 5 == 0:
    ckpt_save_path = ckpt_manager.save()
    print ('Saving checkpoint for epoch {} at {}'.format(epoch+1,
                                                         ckpt_save_path))

  print ('Time taken for epoch {} is {} sec\n'.format(epoch + 1,
                                                      time.time()-start))
```




```
Saving checkpoint for epoch 10 at ./checkpoints/train/ckpt-2
Time taken for epoch 10 is 252.23333621025085 sec
```

Generate using test dataset

```
# Run the trained model on the test dataset
for inp in test_horses.take(5):
  generate_images(generator_g, inp)
```

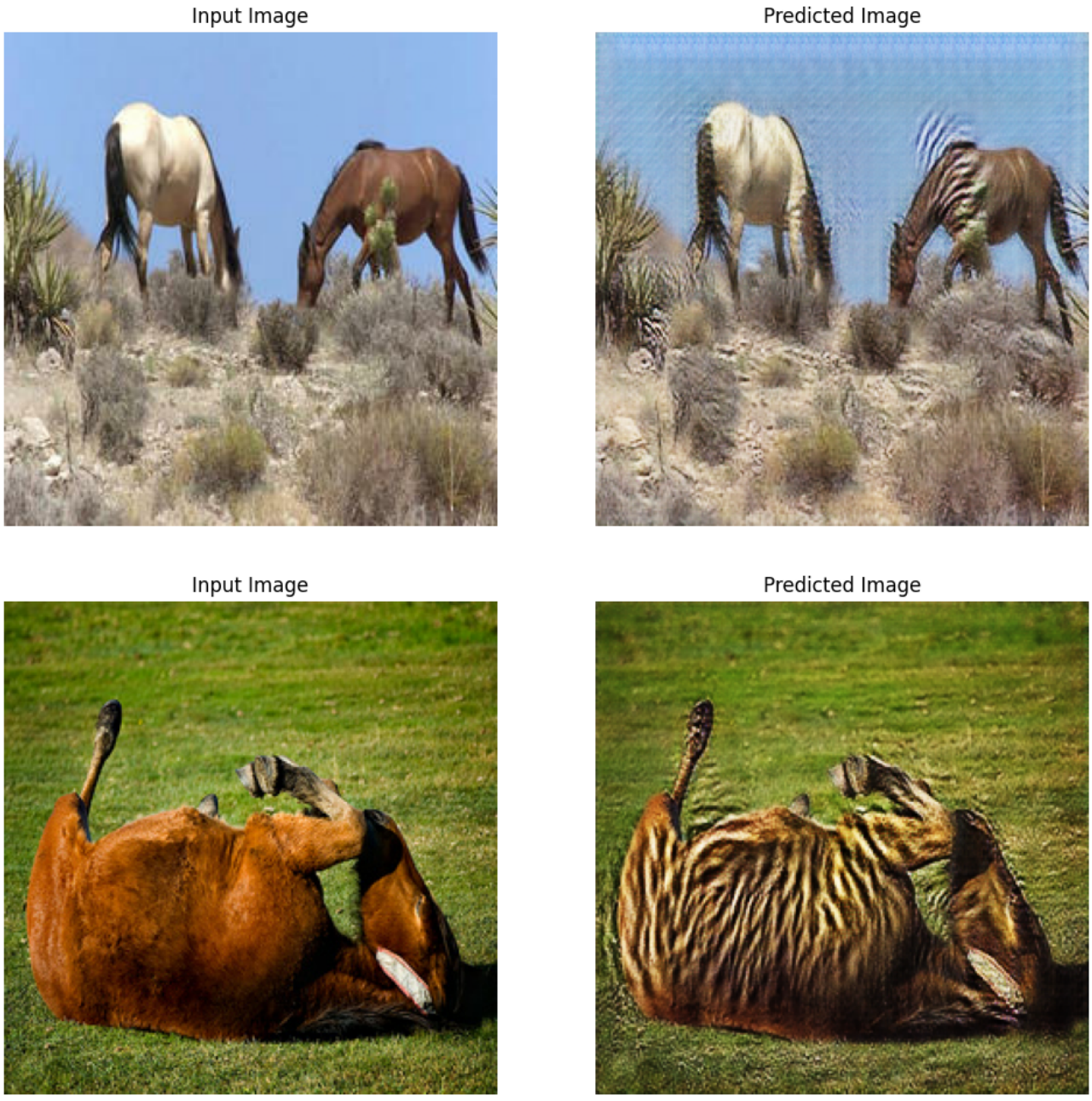


Input Image

Predicted Image

Input Image

Predicted Image



Next steps

This tutorial has shown how to implement CycleGAN starting from the generator and discriminator implemented in the Pix2Pix (https://www.tensorflow.org/tutorials/generative/pix2pix) tutorial. As a next step, you could try using a different dataset from TensorFlow Datasets (https://www.tensorflow.org/datasets/catalog/cycle_gan).

You could also train for a larger number of epochs to improve the results, or you could implement the modified ResNet generator used in the paper (https://arxiv.org/abs/1703.10593) instead of the U-Net generator used here.

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Last updated 2022-12-15 UTC.